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BE (6th Semester)

Examination, April - May, 2016

(Old Scheme)

Communication System - II

Time Allowed : 3 hours *Maximum Marks :* 80

Minimum Pass Marks : 28

Note : (*i*) Part (*a*) of each unit is compulsory. Attempt any **two** parts from (*b*), (*c*) and (*d*).

(*ii*) The figures in the right-hand margin indicate marks.

Unit-I

1. (*a*) What is Aliasing Effect? 2

(b) Find the Nyquist rate and Nyquist interval for the signal 7

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$$x(t) = \frac{1}{2\pi}\cos(4000\pi t)\cos(1000\pi t)$$

(c) Explain the principle of PWM method. Draw the PWN Modulator/Demodulator circuit. 7

(d) Explain T1 digital system which is used to convey multiple signals over telephone lines using wideband coaxial cable. 7

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Unit-II

2. (a) What is granular noise? 2

(b) A μ-law compander is defined as

$$y = \frac{\pm\ln(1+\mu|x|)}{\ln(1+\mu)}, |x|<1$$

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where x is input y is output the + sign is used when x is +ve and – sign is used when x is negative. If the peak of input is 10V and no. of bits available for quantization are then find the smallest μ largest separation between levels. Consider μ = 255. 7

(c) Draw the block schematic of Delta modulation system. How the limitation of DM system is removed by ADM Scheme? 7

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(d) 24 Telephone channels, each band limited to 3.4 kHz are to be time division multiplexed by using PCM. Calculate the bandwidth of the PCM system for 128 quantization levels and an 8 kHz sampling frequency. 7

Unit-III

3. (a) What do mean by bit rate and symbol rate? 2

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(b) For the DPSK system, draw the logic wave form for bit sequence 0010011001111. 7

(c) Draw the block diagram of OQ PSK trasmitter and receiver. Explain the principle of operation of QPSK. 7

(d) Show that MSK has phase continuity for the input binary sequence 1100100010. 7

Unit-IV

4. (a) What is PN sequence? 2

(b) Derive an expression for $\frac{S_o}{N_q}$ ratio calculation of PCM system. (For quantisation noise only). 7

(c) Explain direct sequence spread spectrum modulation technique with suitable diagram. 7

(d) Explain FH-SSM modulation technique. 7

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Unit-V

5. (a) What is Entropy? 2

(b) Find the mutual information for the channel shown 7

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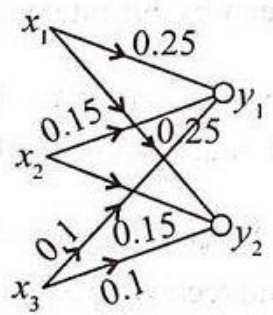

(c) Derive an expression for capacity of a Gaussian channel Shannon-Hartley theorem. 7

(d) Apply Shannon-Fano coding procedure for the following message ensemble : 7

$[x] = [x_1 \quad x_2 \quad x_3 \quad x_4 \quad x_5 \quad x_6 \quad x_7 \quad x_8]$

$[P] = [0.25 \quad 0.125 \quad 0.0625 \quad 0.0625 \quad 0.0625 \quad 0.25 \quad 0.0625 \quad 0.125]$

Take M = 2.

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